OXYPC COMPUTERS · INVENTORY ERP

OxyPC Inventory Lifecycle Design System

Colours, type, layout and components that every page of the app is built from, with the rules for using them.

Version	2.2 — 18 October 2026
Applies to	OxyPC Inventory Lifecycle web app (FastAPI + Jinja2 + Bootstrap 5)
Based on	Code on branch `staging`, `static/css/app.css`, `static/js/global-table.js`
Environments	Production `http://10.199.206.109` · Staging `http://localhost:8001`
Audience	Developers, reviewers and Claude sessions building or changing pages
What's new	v2.2: OxyPC × Bluemonk brand colours (success `#0F7A37`, Bluemonk Navy `#0e3a7a`), new Global Table style, OxyPC green app icon

CONTENTS

What's inside

01	Introduction and principles	Why the system exists, five rules
02	Colour	Brand, semantic, tile and stage colours
03	Typography, spacing and icons	Type scale, radius, shadow, icon set
04	Page layout	Sidebar, top bar, page anatomy
05	Summary tiles	Light-tint, solid, pipeline, stat cards
06	Filter bar	One-row filters, busy-page grid
07	Global Table	Table card, toolbar, behaviours
08	Buttons and badges	Hierarchy, states, stage badges
09	Forms, modals and feedback	Inputs, dialogs, alerts
10	Patterns	Permissions, partial refresh, breadcrumbs
11	Checklist and change process	From idea to production
A	Token and file reference	CSS variables, class names, locations

01

Introduction and principles

Every OxyPC Inventory Lifecycle page is assembled from the same few parts: summary tiles, a filter bar and a Global Table inside a card. This document records how each part looks and how it is built, so that new pages match the 238 templates already live.

The system sits on Bootstrap 5 with a thin shared layer in `static/css/app.css`. Table behaviour lives in `static/js/global-table.js`. All libraries are served locally from `static/vendor/`, with no CDN, so the app works on the office LAN without internet.

Five principles

Principle	What it means in practice
Reuse, don't restyle	Use the shared classes (`filter-bar` , `initGlobalTable` , the tile pattern). A fix in `app.css` then reaches every page at once.
Dense but readable	Tables use 14px text, one-line headers and two-line cell clamps, so 12+ rows fit on a 1366px office screen without wrapping.
Colour carries meaning	Each colour stands for a stage or state and is used the same way on tiles, badges and buttons. Colour is never decoration.
One row of controls	Filters and their buttons sit on one line above the table. Row actions sit in the last column, centred.
Permissions decide what shows	A control appears only for roles that may use it, checked with `has_perm()` or `role_allowed()` . Never a hard-coded role list.

How to use this document

- **Building a new page:** read chapters 04–09, then work through the checklist in chapter 11.
- **Changing an existing page:** find the component's chapter. The samples are rendered with the app's real CSS.
- **Reviewing on staging:** compare the page against the specs and the "Do / Don't" boxes.

> **SOURCE OF TRUTH**
> When this document and the code disagree, the code in production wins. Update this document in the same change — its source is `qa_docs/design_system/design_system.html` ; rebuild the PDF with `build_pdf.py` next to it.

02

Colour

Four groups of colour: the brand colours and neutrals around every page (including the OxyPC greens), Bootstrap's semantic colours for actions and states, tile colours for summary counts, and fixed stage colours for device badges.

Brand and neutrals UPDATED

Bluemonk Navy
`#0e3a7a` --brand-blue
History buttons, table headers, focus rings, sidebar footer, card titles.

Bluemonk Blue
`#0377c2` --sidebar-active
Active sidebar link only

Sky Blue
`#d9ebf6` —
Global Table card header

OxyPC Leaf
`#54BA48` --oxy-leaf
Accents only (logo, icon top) — too light for text

OxyPC Forest
`#129042` --oxy-forest
Profit figures (`.profit-positive`), icon front

Mint
`#E8F5E6` --oxy-mint
Green tile and stat-card background

Light Colour
`#f0f4f8` --sidebar-text
Sidebar link text, and the page background behind all content

Dark Colour
`#212529` body text
Body text, table cells, form fields

Disabled
`#D8D8D8` —
Every disabled button

Semantic colours

Bootstrap 5 defaults, except Success (OxyPC green), plus three custom badge colours defined in `app.css` (teal, purple, pink).

Primary
`#0d6efd` primary
Main actions, QC Check, Active

Success UPDATED
`#0F7A37` success
Save, Ready to Sell, done.

Warning
`#ffc107` warning
GRN, L1/L2, Returned, pending

Danger
`#dc3545` danger
L3, Scrapped, PNA, delete

Info
`#0dcaf0` info
Stock In, TRC Production

Secondary
`#6c757d` secondary
IQC, Inactive, neutral

Dark
`#212529` dark
Sold, Scrap for Sale

Teal
`#0d9488` bg-teal
All cosmetic stages

Purple
`#6f42c1` bg-purple
Final QC, "at you" counts

Pink
`#d63384` bg-pink
Cosmetic summaries

Tile colours UPDATED

Summary tiles set one variable, `--tc`, and derive their tint, border, number and watermark from it. These are the shades in use. They are darker than Bootstrap's warning and info, which are too pale for coloured text on a light tint. Green tiles sit on mint `#E8F5E6`.

Blue
`#0d6efd`
IQC, L1/L2, Active

Green
`#0F7A37`
Stock, Final QC, Done

Orange
`#fd7e14`
TRC, L3/L4

Amber
`#f0ad00`
Price pending (solid tiles)

Teal
`#0aa2c0`
Ready to Sell, parts

Mint
`#20c997`
Stress

Red
`#dc3545`
PNA, failures

Pink
`#e83e8c`
Cosmetic

Purple
`#6f42c1`
At you

Grey
`#6c757d`
Returned, Inactive

Device stage colours

Fixed in `STAGE_COLORS` (`models/device.py`) and read through `device.stage_color`. Never choose a stage colour by hand.

GRN · IQC · Stock In · L1 · L2 · L3 · TRC Production · QC Check · Cosmetic Received · Cleaning · Painting · Cosmetic Completed · Final QC · Final QC Pass Hold · Final QC Fail Hold · Ready to Sale · Sold · Returned · Scrapped · Scrap for Sale

Rendered with the app's badge classes. All cosmetic sub-stages (Putty, Dry Sanding, Masking, Water Sanding) are teal. Success-green stages show the v2.2 solid `#0F7A37` badge.

Money and profit UPDATED

Class	Colour	Use
`.profit-positive`	#129042 ₹12,450	Profit, positive P&L (was #0f9e4e)
`.profit-negative`	#d63031 −₹3,200	Loss
`.profit-zero`	#888 ₹0	Break-even

DO
Pick a tile or badge colour from what it counts: green for stock and done, orange for repair, red for failures.

DON'T
Use a `bg-*` class that isn't in Bootstrap or `app.css`. The badge renders white text on no background.

03

Typography, spacing and icons

The app uses Segoe UI (Windows), falling back to Arial. The whole interface renders at 90% zoom, so the sizes below are what the CSS says, before that zoom.

Type scale

Role	Spec	Sample
Page title (top bar)	1.25rem · bold · primary blue	Inventory Manager
Tile number	1.6rem · 700 · tile colour	3,518
Card title	1rem · semibold · navy	Inventory Stock
Body / table text	14px · 400	LOT-042 · Dell Latitude 5490 · i5 8th Gen
Table header	0.8rem · 600 · uppercase · 0.5px tracking	TAG NUMBER
Form label / sidebar link	0.875rem	Device Type
Tile label	0.72rem · #343a40	Devices in Stock
Sidebar section	0.7rem · 700 · uppercase · 1px tracking	INVENTORY
Barcode field	Courier New · 1rem	OXY-2026-004512

Spacing, corners and shadow

Token	Value	Where
Card and tile radius	10px	Cards, summary tiles, `.stat-card`
Table radius	0.5em	Tables inside cards (clip-path on scroll tables)
Control radius	6px	Inputs, DataTables search and length controls, sidebar links
Grid gap	`g-3` = 1rem	Between tiles and cards
Below tile row	`mb-4` = 1.5rem	Tiles → filter bar
Shadow	`0 .25rem .5rem rgba(0,0,0,.5)`	`.shadow-sm` is overridden app-wide to this. Use the class, don't write your own
Cell padding	8px (`.5rem`); first column 16px left NEW	Every Global Table, even when `table-sm` is set
Sidebar / top bar	240px / 56px	`--sidebar-width`, `--topbar-height`
App zoom	0.9	`--app-zoom`. Use `var(--full-vh)` instead of `100vh`

Icons

Bootstrap Icons only, as `<i class="bi bi-…">`. Use `me-1` before button text and `me-2` before sidebar and card titles. Reuse the icon a module already has in the sidebar.

Stock Inwards `bi-archive`	Production `bi-cpu`	IQC `bi-clipboard-check`	In stock `bi-box-seam`
Repair / TRC `bi-tools`	Ready to sell `bi-bag-check`	Filter `bi-funnel`	Download `bi-file-earmark-arrow-down`
Upload `bi-upload`	Edit `bi-pencil`	Delete / scrap `bi-trash3`	Done `bi-check-circle`
Warning / PNA `bi-exclamation-triangle`	Scan `bi-upc-scan`	Assigned to you `bi-person-workspace`	Transfer `bi-arrow-left-right`

App icon NEW

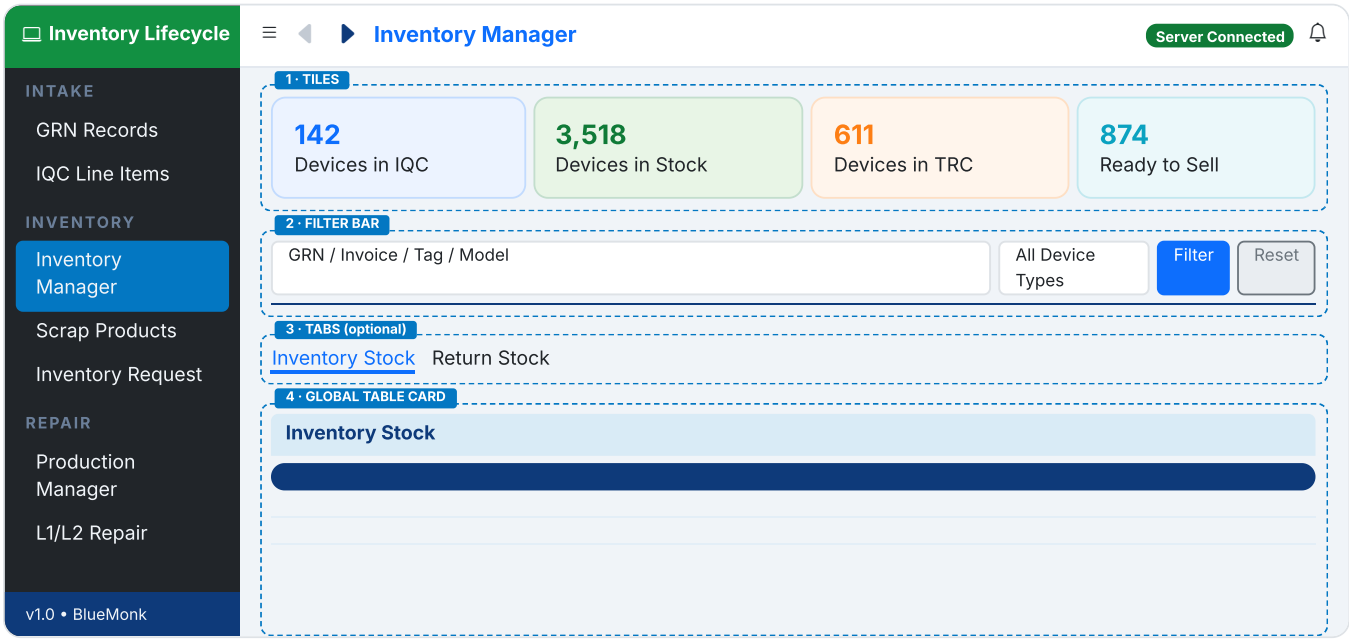


A 3D stock box in the OxyPC greens (top `#54BA48`, front `#129042`, side `#0B6B30`) with a delivery van on the lid, a laptop on the front and a white tick. `static/img/logo.svg` (dark rounded tile, browser-tab icon) and `logo-mark.svg` (no tile, login badge). The sidebar header is unchanged.

04

Page layout

Every page extends `base.html`, which provides the dark sidebar and white top bar. A page fills only its `content` block, plus `head` for page-specific CSS.



Page anatomy: the standard order of a list page, top to bottom.

Standard order of a list page

#	Zone	Rule
1	Summary tiles	Optional. When present, always first, above any tabs. One row.
2	Filter bar	One line of filters, Filter/Reset on the right, Bluemonk Navy line underneath.
3	Alert	Only after a redirect with `?success=` or `?error=`.
4	Tabs	`nav nav-tabs mb-3` when the page has several tables. A table inside a tab hides its own card header.
5	Table card	Sky Blue card header with icon + title and this table's own actions, then the Global Table.

Sidebar and top bar UPDATED

- **Sidebar colours:** brand band OxyPC Forest `#129042` reading "Inventory Lifecycle"; link text `#f0f4f8` (the page background); footer in Bluemonk Navy; active link `#0377c2` (`--sidebar-active`). Green icons and badges in the sidebar use the same success green `#0F7A37` as the rest of the app.
- **Back / forward:** filled triangles (`bi-caret-left-fill` / `-right-fill`) in Bluemonk Navy `#0e3a7a` , pale grey `#c3cad3` when there is nowhere to go; no pill or background.
- **Sections:** Attendance, Intake, Inventory, Repair, Store Manager, CRM, External Partner, Customer Care, Sales & CRM, Finance, Product Master, Reports, Admin, Notifications, Support.
- **Each link** is wrapped in `has_perm(role,'<module>','enable') and not module_hidden('<module>')` . Its text comes from `sidebar_label()` , so admins can rename it in Sidebar Config.
- **Page title** comes from the `page_title` block and can be overridden in Page Title Config. Breadcrumbs are off by default and switched on per page there.
- **Widths:** layouts must hold at 1366px. Below 768px, the sidebar becomes a slide-in drawer.

05

Summary tiles

Four tile styles. Light-tint is the standard for count tiles on working pages. Solid tiles are for status and category overviews. The pipeline and stat cards belong to the Dashboard.

A · Light-tint tile (standard)



Production Manager (`/trc-production`). The same tile is used on Inventory Manager (`/stock`) and Create GRN (`/grn/post-iqc`).

Part	Spec
Background	8% of the tile colour over white: `color-mix(in srgb, var(--tc) 8%, #fff)` ; green tiles use mint `#E8F5E6` . One variable, `--tc` , set per tile.
Box	1px border at 25% of the tile colour · radius 10px · padding `.6rem .7rem` · shadow `0 .25rem .5rem rgba(0,0,0,.5)`
Text	Number 1.6rem, weight 700, tile colour · label .72rem, `#343a40` , one line, **.35rem below the number**
Watermark	Bootstrap icon, 3.4rem, bottom-right (`right:-4px; bottom:-10px`), **8% of the tile colour**, partly cut off
Optional button	`btn-sm py-0 px-1` , .65rem, **.6rem below the label**; white with tile-colour border and text; fills with the tile colour on hover
Row	`row g-3 mb-4` , padding `4px 6px 10px` so the shadow isn't clipped. `col-6 col-lg-3` for 4 tiles; 5–8 tiles use a no-wrap flex row (`d-flex gap-3` , min-width 118px)

```
<div class="prod-tile h-100" style="--tc:#0F7A37">
  <div class="prod-tile-count">{{ devices_in_stock }}</div>
  <div class="prod-tile-label">Devices in Stock</div>
  <i class="bi bi-box-seam prod-tile-wm"></i>
</div>
```

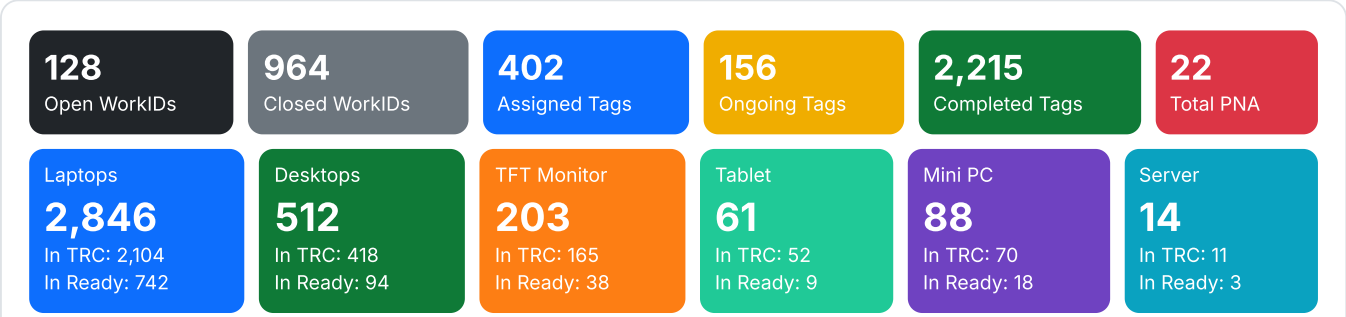
DO

One tile, one count, with a label that names it ("Total Tags in L1/L2"). Tiles go above the filter bar. A count tile that leads somewhere gets a small Summary button.

DON'T

Make a count tile itself a link, or put tiles below the filter bar. Whole-tile links are only for the Dashboard's category cards and pipeline (B and C).

B · Solid tile (status and category overviews)



Top: Product Master and WorkID Status (`.pm-tile` , `col-xl-2`). Bottom: Dashboard category cards (`.dash-tile`), each linking to All Inventory for that category. Green is the v2.2 success green `#0F7A37` .

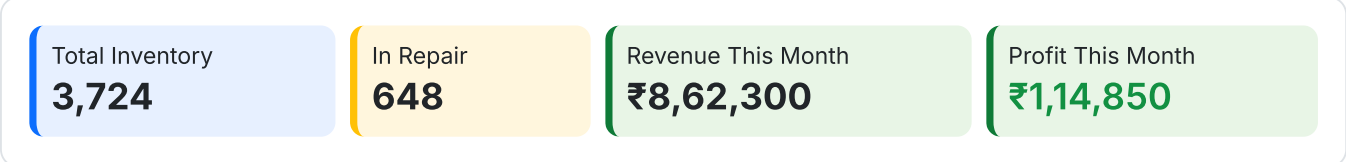
Part	Spec
Count tile	Solid fill, `p-3` , radius 10px, min-height 96px, shadow `0 1px 3px rgba(0,0,0,.08)` . White number 1.9rem bold; label .82rem at 88% white; watermark 4.6rem at 22% white.
Category card	Same, but min-height 120px: label first (.9rem, semibold), number 2.2rem, then two .8rem sub-lines (In TRC / In Ready); watermark 5rem. Lifts 4px with a larger shadow on hover.

C · Pipeline tile (Dashboard only)



Part	Spec
Tile	White, `rounded-3` , min-width 80px, shadow `0 .125rem .35rem rgba(0,0,0,.15)` . Coloured title strip on top (`bg-{stage colour}` , .75rem, weight 600); count below: 1.25rem bold.
Row and card	All ten stages on one row (`flex-wrap:nowrap`) with a filled green arrow between them (`bi-caret-right-fill text-success` , now #0F7A37). Split Entity / Split Zone as white outline buttons; Bluemonk Navy watermark 7rem at 6%.

D · Stat card (Dashboard only)

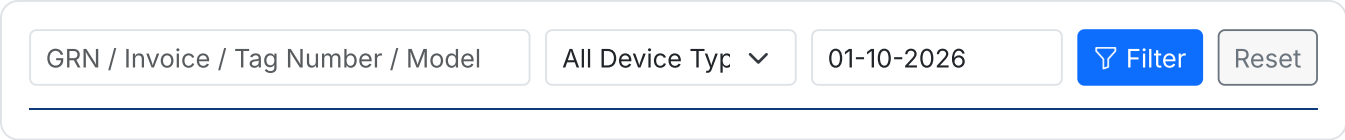


Part	Spec
Card	Light tint with a 4px coloured left edge: `stat-primary` #e7f0ff, `stat-warning` #fff6dd, `stat-success` **#E8F5E6** with edge #0F7A37, `stat-danger` #fdeaea.
Money	₹ with Indian grouping. Profit This Month switches to `stat-danger` and red text when negative.

06

Filter bar

One row of small controls sitting on the page, with no box, and a Bluemonk Navy rule underneath. Every list page uses it.



Rendered with the app's `.filter-bar` CSS.

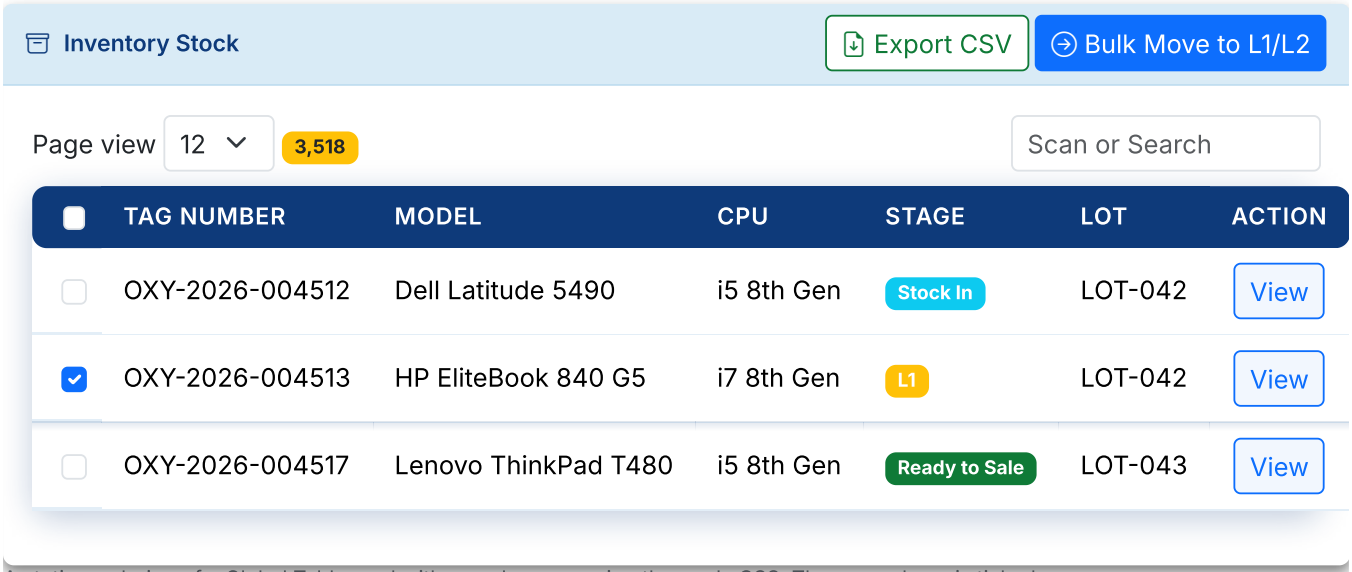
```
<form method="get" action="/stock" class="filter-bar">
  <div class="filter-bar-row">
    <div class="filter-bar-search"><input class="form-control form-control-sm" title="Search" placeholder="GRN
/ Invoice / Tag / Model"></div>
    <div><select name="device_type" class="form-select form-select-sm" title="Device Type">…</select></div>
    <div><input type="date" name="date_from" class="form-control form-control-sm" title="Date From"></div>
    <div>
      <button class="btn btn-primary btn-sm"><i class="bi bi-funnel me-1"></i>Filter</button>
      <a href="/stock" class="btn btn-outline-secondary btn-sm">Reset</a>
    </div>
  </div>
</form>
```

Rule	Detail
One row	Fields share the width and shrink together (min 70px). The last item, the buttons, sits at the right edge.
Search box	`filter-bar-search` takes a double share (min 120px).
Checkbox	`filter-bar-check` keeps its natural width so its label never truncates.
No labels	Each field has a `title=` tooltip and a placeholder or "All ..." first option.
Fields	`-sm` sizes; white background, black text, including multi-select (`ms-filter`).
Busy pages	More than ~6 filters: `filter-bar filter-bar-6` , a 6-column grid that wraps; buttons share the last column.

07

Global Table UPDATED

`initGlobalTable('#id', dtOptions, opts)` wraps a server-side DataTable with the app's shared behaviour and look. It's used on 40+ pages. Never hand-build a list table.



A static rendering of a Global Table card with sample rows, using the app's CSS. The second row is ticked.

Built-in behaviour

Behaviour	Default
Look NEW	Bluemonk Navy `#0e3a7a` header with white labels and rounded ends; plain white rows with a thin `#e3eef7` divider; soft Bluemonk Navy shadow behind the table. Added automatically (`.gtable-cards` on the DataTables wrapper); `opts.plainStyle:true` opts out.
Selection and hover NEW	Ticked row: pale blue `#eef6fc`, blue left edge, light shadow. Row under the mouse: deeper shadow, always above a ticked row.
Header separators NEW	Thin white line (25%) between header labels; 12px left/right padding in header and body cells (first column keeps 16px left)
Sorting NEW	Sort arrows sit 12px after the label (soft white; the active direction turns bright white). Every data column sorts, on the server for server-side tables: dates by date, numbers by number, text A–Z, Stage in workflow order, empty values last. Action columns (header "Action"/"Actions" or blank) never sort. Lists, counts worked out per row and button columns stay unsortable.
Rows per page	12
Text	14px; headers one line, uppercase; columns size around their header
Long cells UPDATED	Cells stay on one line (the table scrolls sideways instead of squashing). 32+ characters clamp to 2 lines within 160–260px; full value in a hover tooltip (`title=`)
Wide tables	Scroll sideways inside the card; first column frozen (first two when column 1 is a checkbox)
Toolbar	Left: "Page view" dropdown + yellow row-count badge. Right: search / scan box + pagination
First / last column	First has 16px left padding, left-aligned (centred only when a checkbox); last is Actions, centred, never wraps
Footer total row	Grey `#e9ecef` (disabled colour), so it reads as a computed summary
Empty state	Built-in image and caption. Never hand-write a placeholder `<tr>`, which crashes DataTables
Options	`opts.selectAll` header checkbox; `opts.scan` barcode scan-to-select; `opts.freeze:false` to opt out of scrolling

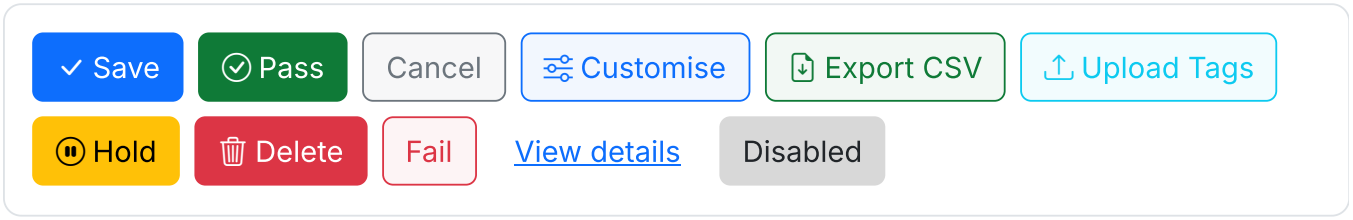
Card rules

- **Header:** `card-header bg-transparent d-flex justify-content-between align-items-center`. It turns Sky Blue `#d9ebf6` with a Bluemonk Navy title automatically.
- **Left:** icon + title. **Right:** only this table's own actions or filters. Never a count badge, because the toolbar already shows the count.
- **Header row:** styled by `initGlobalTable` (Bluemonk Navy, rounded); `table-light` for small static tables in forms and detail pages.
- A last column that holds text rather than buttons: add `gtable-last-col-text`.
- After editing `global-table.js`, keep it in `_VERSIONED_ASSETS` (`templates_config.py`) so browsers fetch the new copy.

08

Buttons and badges

Button hierarchy



Rendered with the app's CSS, including its outline-button tint, grey disabled style and the v2.2 success green `#0F7A37`.

Use	Class	Examples
Main action	`btn btn-primary`	Save, Filter, Create Lot
Positive step	`btn btn-success` (#0F7A37, hover #0B6B30)	Pass, Move to Ready, Approve, Sell
Secondary / cancel	`btn btn-outline-secondary`	Reset, Cancel, Back
Supporting actions	`btn-outline-primary / -success / -info`	Export, Upload, Customise
Needs attention	`btn btn-warning`	Hold, Return
Destructive	`btn-danger` / `btn-outline-danger`	Delete, Fail, Scrap. Always confirm first
Inline link	`btn btn-link`	View details

- **Size:** `btn-sm` everywhere except the submit button at the bottom of a full form.
- **Labels** are a verb + object ("Move to Final QC"), never wrap, and have an icon with `me-1`. Icon-only buttons need `title=`.
- **Outline buttons** sit on a 5% tint of their colour and fill solid on hover (app-wide).
- **Disabled** is always grey `#D8D8D8` with dark text. Don't restyle it.
- **Destructive** actions confirm with the count: `return confirm('Delete 3 lots?')`.

Badges UPDATED

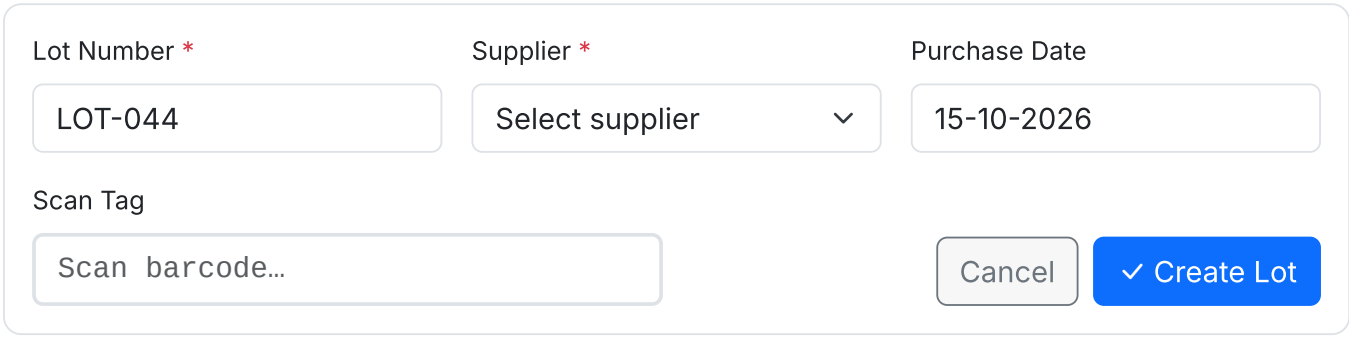


Badge	Rule
Stage	`badge bg-{{ device.stage_color }}` + `stage_label`; never chosen by hand
Success	`badge bg-success`: solid **#0F7A37** with white text (v2.2)
Status	Done/active green · pending yellow with `text-dark` · failed red · inactive grey
Notification count	`badge rounded-pill bg-danger`
Row count	Yellow toolbar badge from the Global Table; not repeated in the card header

09

Forms, modals and feedback

Forms



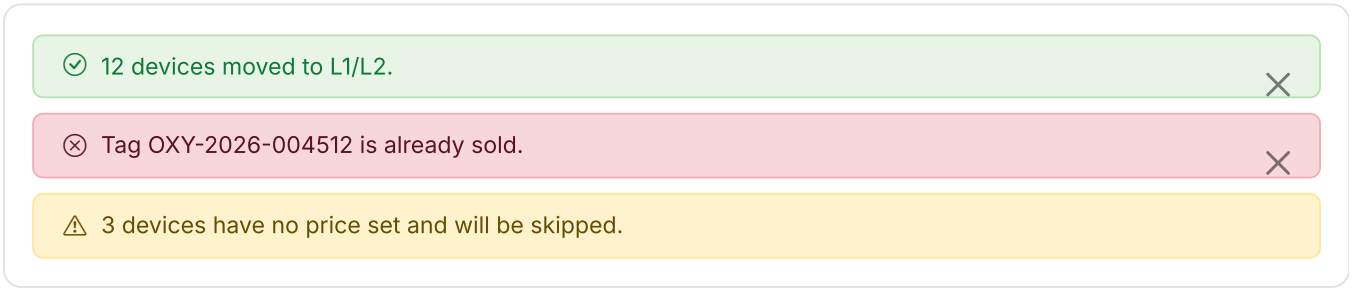
- `form-label` (0.875rem) above each field; required fields marked `<span class="text-danger">*</span>`.
- Fields in a Bootstrap grid (`row g-3` , `col-md-4`), grouped in cards by topic. `-sm` sizes inside filter bars, tables and modals.
- Barcode and tag inputs use `barcode-field` (Bluemonk Navy focus ring) and get focus on page load for scanners. The login page is the exception: no field auto-focuses.
- Money shows as ₹ with Indian grouping. Totals are calculated on the server, never in the browser (skill `server-side-financial-snapshot`).

Modals

All modals use the same frame: a white `modal-header` with an `h5.modal-title` (icon + title) and a close button, a `modal-body` , and a `modal-footer` with Cancel on the left of the main button.

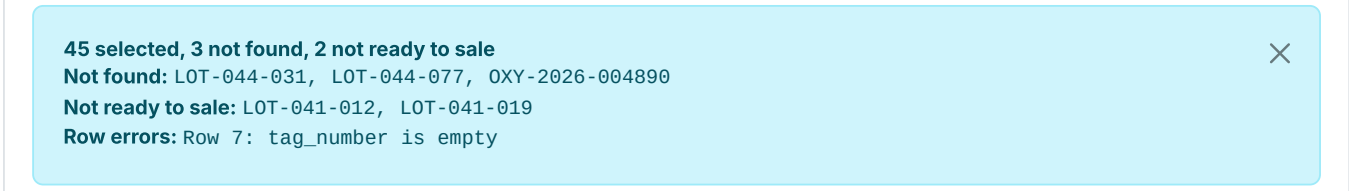
Part	Rule
Size	Default (500px) for most forms · `modal-sm` for one field or a short confirmation · `modal-lg` for tables and long forms · add `modal-dialog-scrollable` when content can grow, `modal-dialog-centered` for confirmations
Header	White. `<h5 class="modal-title">` with a Bootstrap icon and `me-2` , then `btn-close` . The title names the action ("Assign Bucket"), not the page
Tinted header	Only to signal the type: `bg-primary bg-opacity-10` informational · `bg-warning bg-opacity-10` caution · `bg-danger bg-opacity-10` + `modal-title text-danger` destructive or defect
Body — fields	Context line first (`small text-muted` , e.g. "3 tag(s) selected"). Labels `form-label small fw-semibold` , required `*` in `text-danger` . Controls `-sm` ; two short fields side by side in `row g-2`
Body — no fields	Summary rows: label `text-muted small` + value in `<strong>` , so the user sees exactly what will happen before confirming
Errors	An `alert alert-danger small` at the top of the body, hidden (`d-none`) until the server returns an error. The modal stays open, the fields keep their values, the button is re-enabled. Mark the bad field `is-invalid` with `invalid-feedback`
Footer	Cancel `btn btn-outline-secondary` (`data-bs-dismiss="modal"`) left of the main `btn btn-primary` with icon. A secondary destructive action sits far left with `me-auto` . Informational modals have one OK/Close button
Behaviour	Disable the main button while saving. On success, close the modal and refresh the table (`ajax-partial-refresh`) or redirect with `?success=`

Alerts and feedback



- After a form submit, redirect with `?success=` or `?error=`; the page shows a dismissible alert above its content.
- `alert-warning` for what the user must know before acting; `alert-info` for neutral guidance. One sentence, with the number.
- No alert for "no data": the Global Table's empty state covers it.
- **Close button.** Page alerts use `py-2`; add `.alert-dismissible.py-2 .btn-close { padding: .75rem 1rem; }` to `app.css` to centre the ×.

Upload result banner



- Shown after **Bulk Upload Tags** on Ready to Sale (and other pages that tick rows from a CSV). The modal closes, the banner is added at the top of `<main>` and the page scrolls to it.
- `alert alert-info alert-dismissible small mb-3`. Bold summary line with every count, then one line per problem group with the tags in `font-monospace`.
- The one place an alert runs past one sentence: the user needs the exact tags to fix the file. It never auto-hides.

Upload results and errors

Bulk Lot, Device (IQC) and Spare Parts uploads save valid rows, skip and list bad ones, and review duplicates.

Part	Rule
Count alerts	Two side by side (`col-md-6`), not dismissible: rows inserted in `alert-success`; rows with errors in `alert-warning` if any, else `alert-success`.
Error table	`table table-sm table-bordered small`, `table-warning` header, columns # and Error. Each message starts `Row N:` and names the field and bad value.
Actions	Back to List (`btn-primary`), Upload Another File (`btn-outline-secondary`).
Duplicate review	Opens by itself for repeated Tag Numbers or unknown lots. `modal-xl`, static backdrop; Update All and Skip and Submit left of the close button. Row actions are small outline buttons: Approve Move (primary), Update (warning), Add Lot (success).

10

Patterns

Showing and hiding by permission

A control the user can't use is hidden, not disabled. A control they can use must always render, including for custom roles that admins create (for example `trc_manager`).

Need	Use
Module, sidebar link, section	`has_perm(role,'<module>','enable')` + `not module_hidden('<module>')`; `any_perm()` for a section header
Role-specific button	`role_allowed(current_user.role, 'admin', 'sales_manager')`. It mirrors the server's `require_roles()` and lets custom roles through
Server side	Always re-checked with `require_roles` and `verify_csrf`. Hiding is for clarity, not security

```
{% if role_allowed(current_user.role,
      'admin', 'sales_manager') %}
  <button class="btn btn-success btn-sm">
    Approve</button>
{% endif %}
```

DON'T

`current_user.role.value in ['admin','sales_manager']`. Custom roles silently lose the button while the server would allow them.

Updating part of a page

When an action should refresh only a table or the tiles (for example a bulk assign in a modal):

1. The POST endpoint returns JSON when the request has `X-Requested-With: XMLHttpRequest`, and still redirects otherwise.
2. The page submits with `fetch()` + `FormData`, closes the modal, then calls `dt.ajax.reload(null, false)` (keeps the current page) or swaps the region by ID with `DOMParser`.
3. Re-attach event handlers in swapped HTML and refresh tile counts.
4. Without `fetch`, the normal form submit still works.

Page titles and breadcrumbs

- Admins rename any page in **Admin → Page Title Config**; the top bar shows the custom title, otherwise the template's `page_title`.
- Breadcrumbs are **off by default** (since 15 Oct 2026) and switched on per page in the same screen.

11

Checklist and change process

Design checklist for a new or changed page

✓	Check
☐	Extends `base.html`; sets `title` and `page_title`; sidebar link gated by `has_perm` with a `sidebar_label`
☐	Order: tiles → filter bar → alert → tabs → table card
☐	Tiles use the light-tint spec, with colours from the tile palette
☐	Filters use `filter-bar`: one row, `-sm` controls, `title=` instead of labels, Filter/Reset on the right
☐	Table uses `initGlobalTable` in a card with icon + title; actions in the last column
☐	Buttons follow the hierarchy; destructive ones confirm with a count
☐	Stage badges use `stage_color`; no unknown `bg-*` classes
☐	Greens use the OxyPC tokens (success `#0F7A37`), never Bootstrap's old #198754
☐	Role checks use `role_allowed()` / `has_perm()`, never a hard-coded list
☐	Works at 1366px; wide tables scroll inside the card
☐	No CDN links; any new library goes in `static/vendor/`

From change to production

#	Step	What happens
1	Database approval	A new table or column is drafted as `.dbml` and approved in dbdiagram.io before it is built.
2	Build and verify	Templates must parse. Run skill `verify-feature`; run `regression-sweep` if `base.html`, `app.css`, a macro or a Jinja global changed.
3	Staging	Changes go to the `staging` branch. Staging machines (Windows `RUN_STAGING.bat`, Mac `START_STAGING.command`) load it at `http://localhost:8001`; check with Ctrl+F5.
4	Approval	Pushed only after an explicit "push" for that batch. An earlier yes never carries over (`commit-deploy-gate`).
5	Production	Merge to GitHub `main` (`Bluemonk123/oxypc-inventory`). Production at `http://10.199.206.109` updates within 2–5 minutes.
6	Keep this document current	Edit `qa_docs/design_system/design_system.html` in the same change and rebuild the PDF.

APPENDIX A

Token and file reference

CSS variables (`app.css :root`) UPDATED

Variable	Value	Purpose
`--sidebar-width`	240px	Sidebar width
`--sidebar-text`	#f0f4f8	Sidebar link text (same as page background)
`--sidebar-active`	#0377c2	Active sidebar link
`--brand-blue`	#0e3a7a	History buttons, focus, brand (was #0377c2)
`--oxy-leaf` / `--oxy-forest`	#54BA48 / #129042	OxyPC greens: accents / profit
`--oxy-text` / `--bs-success`	#0F7A37	Success: buttons, badges, green text, tiles
`--oxy-mint`	#E8F5E6	Green tile / stat-card background
`--topbar-height`	56px	Top bar height
`--app-zoom`	0.9	Global UI scale
`--full-vh`	calc(100vh / var(--app-zoom))	Use instead of 100vh

Shared classes

Class	What it does
`filter-bar` , `filter-bar-row` , `filter-bar-search` , `filter-bar-check` , `filter-bar-6`	Filter bar layout
`gtable` , `gtable-card` , `gtable-top` , `gtable-cards`	Global Table (added by `initGlobalTable`); `gtable-cards` = Bluemonk Navy header, row shadows
`gtable-last-col-text`	Last column is text: left-aligned and wraps
`gtable-clamp2`	Two-line clamp with tooltip, 160–260px wide
`bg-teal` , `bg-purple` , `bg-pink`	Custom badge colours
`barcode-field`	Monospace scan input with Bluemonk Navy focus ring
`card-hover`	Lift-on-hover for clickable KPI cards
`profit-positive` / `-negative` / `-zero`	P&L colours (#129042 / #d63031 / #888)
`prod-tile*` , `inv-tile*` , `pm-tile*`	Page-level tile classes (light-tint, light-tint, solid)

Jinja globals for UI

Global	Use
`has_perm(role, module, action)`	Module permission check
`any_perm(role, *modules)`	Show a section if any module is allowed
`role_allowed(role, *roles)`	Role check that is safe for custom roles
`module_hidden(key)`	Master Data global visibility
`sidebar_label(key, default)`	Admin-renamable sidebar text
`resolve_page_title(path)` , `breadcrumb_enabled(path)`	Page Title Config
`app_version()`	Version shown on the login page and sidebar footer

Where things live

Path	Contents
`templates/base.html`	Sidebar, top bar, alerts, breadcrumbs, script includes
`static/css/app.css`	All shared overrides, colour tokens

`static/js/global-table.js`	`initGlobalTable`
`static/img/logo.svg` , `logo-mark.svg`	App icon (browser tab, login badge)
`static/vendor/`	Bootstrap 5, Bootstrap Icons, DataTables, Chart.js (local copies)
`templates_config.py`	Jinja globals, asset versioning
`models/device.py`	`STAGE_COLORS`
`qa_docs/design_system/`	This document's source (`design_system.html`) and PDF builder
`.claude/skills/`	Project workflows: verify-feature, regression-sweep, commit-deploy-gate, custom-role-permissions, ajax-partial-refresh